Catalysis and Catalytic Reactors. III

Catalytic Reactors & Catalysis Deactivation:

Connecting Theory to Real World

CENG252
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Catalytic Reactors

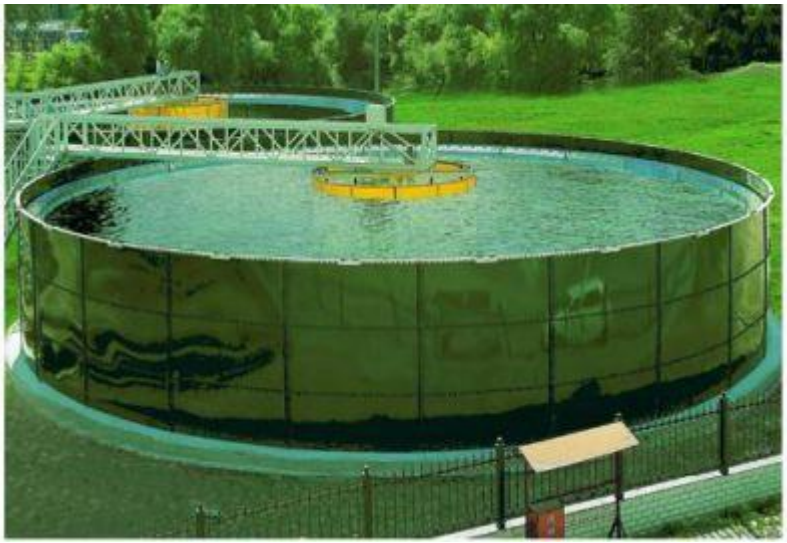
- Continuous Stirred Tank Reactor
- Moving Bed Reactor
- Packed Bed Reactor
- Straight-Through Transport Reactor (STTR)

CSTRs

PROJECT REFERENCE

Beijing Olympics Games Water Treatment Project

4 sets of sedimentation tank Ø18.33×3.6M
1 set of mud collection tank Ø4.82×3.6M
Total volume: 5200m3, Installation: 12 person, 15 days







Stacked CSTR



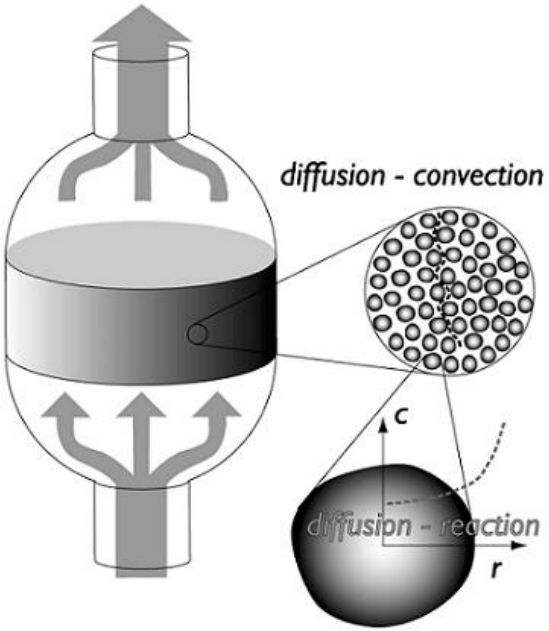
The unit shown in this photo is a stacked reactor platform, which (in this case) is a set of four reactors in series. The reactors are stacked on top of one another to minimize the amount of ground space required for the reactors.

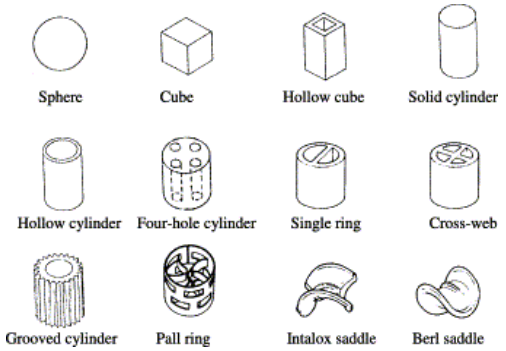
Because this reaction requires high temperatures, the furnace at the left of the photo is used to heat the initial feed material to around 1000° F. The reaction is also endothermic, which means the material cools as it passes through each reactor, and must be re-heated back up to proper reaction temperature between each pair of reactors. The large silver (insulated) pipes on the left side of the reactors transport material through the furnace between each pair of reactors

Packed Bed Reactor



PBR **reactors** are tubular and are filled with solid catalyst particles, most often used to catalyze gas reactions. The chemical reaction takes place on the surface of the catalyst. The advantage of using a **PBR** is the higher conversion per weight of catalyst than other catalytic **reactors**.



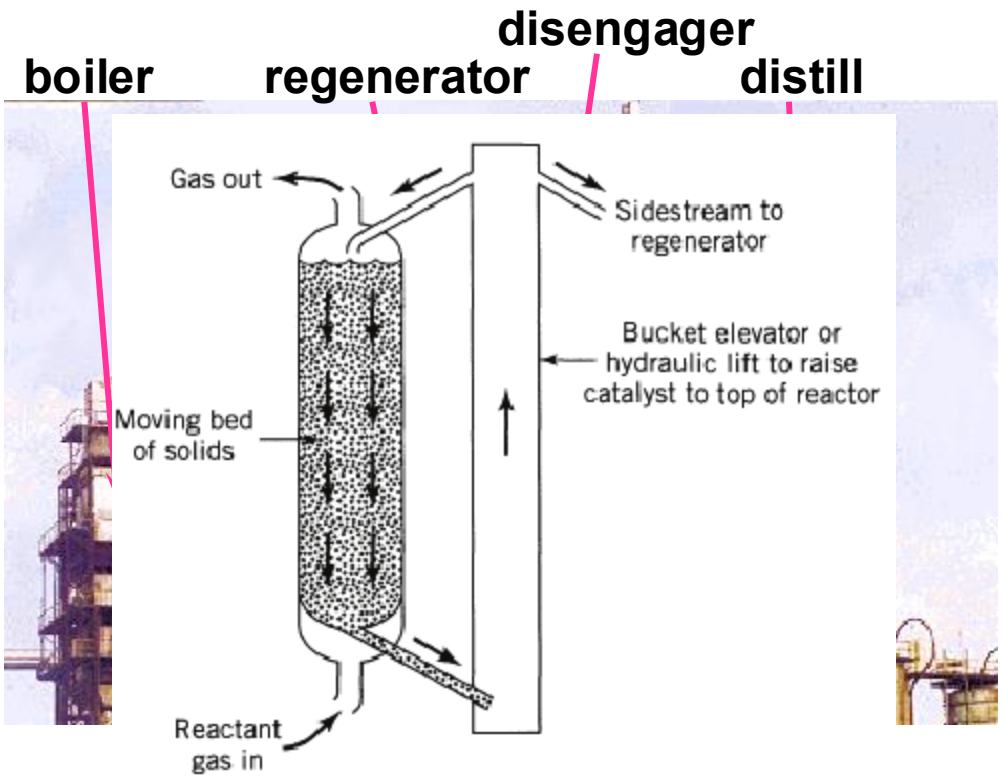




- Synthesis gas production
- Methanol synthesis
- Ammonia synthesis
- Fischer-Tropsch synthesis
- Gas cleaning (adsorption)

Moving Bed Reactor

MBRs are reactors in which the catalytic material flows along with the reactants and is then separated from the exit stream and recycled. (cracking, waste water etc.)



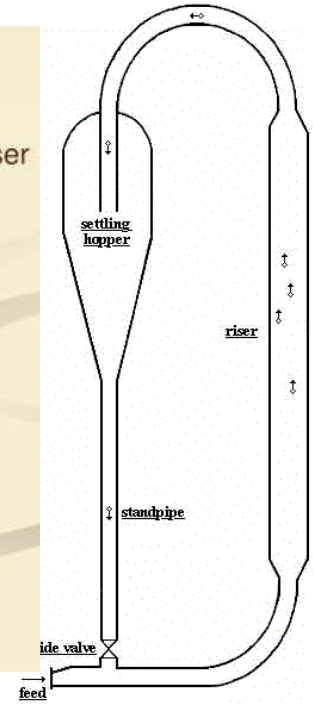
This photo is of a catalytic cracker moving bed reactor. Only the disengager section of the reactor is visible in the picture - the reaction section is hidden behind the stair structure beneath the disengager.

The cracker is used for the catalytic cracking of gas oil into light aromatics and straight chain hydrocarbons, which are then separated in the distillation tower to the right of the photo.

The white unit to the left of the cat cracker is the catalyst regenerator, where coke deposits are burned off the catalyst. This is a highly exothermic operation, (1400° F), and so the boiler unit at the far left recovers the sensible heat of the regenerator exhaust to produce steam.

Straight Through Transport Reactor (STTR)





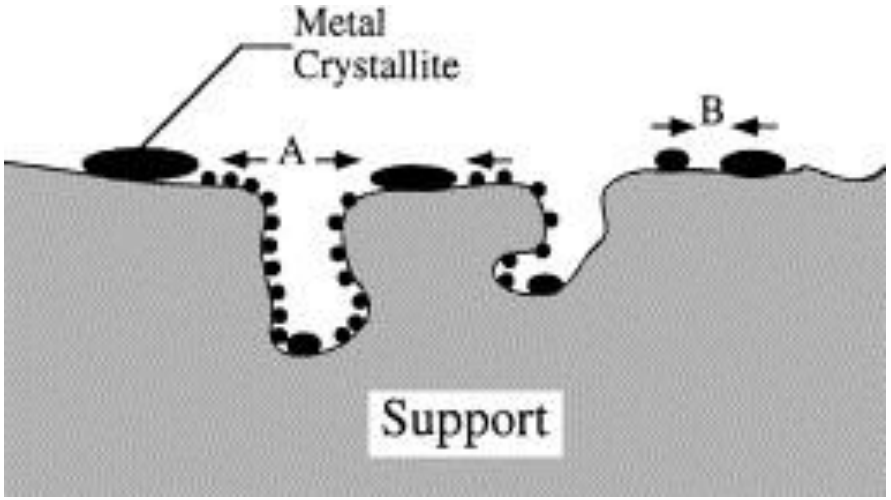
Straight Through Transport Reactor (STTR) in use for a Fisher-Tropsch synthesis reaction at Sasol Limited Chemical.

$$(2n + 1)\ H_2 + n\ CO \rightarrow C_nH_{2n+2} + n\ H_2O$$

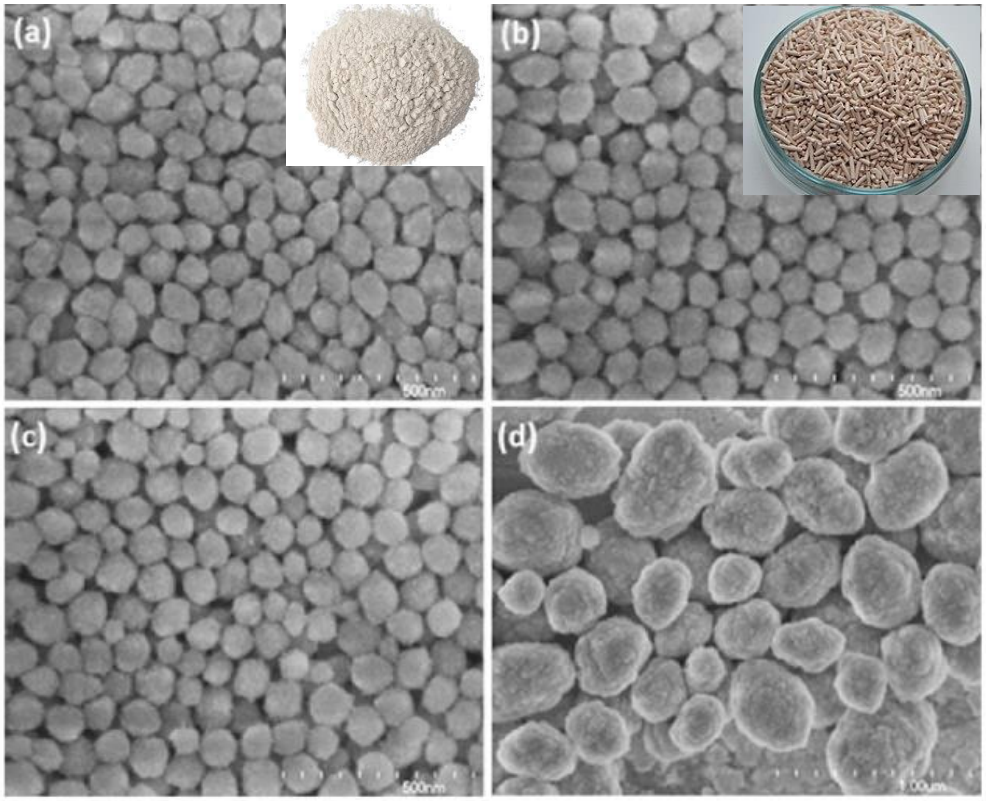
STTRs are used widely in the production of gasoline from heavier petroleum fractions . In addition, it has found use in grain drying operations. In this reactor, either an inert gas or the reactant itself transports the catalyst through the reactor. With this reactor, any possibility of catalyst decay/selectivity disguise is virtually eliminated because the catalyst and reactants are fed continuously.

Functional Catalysts

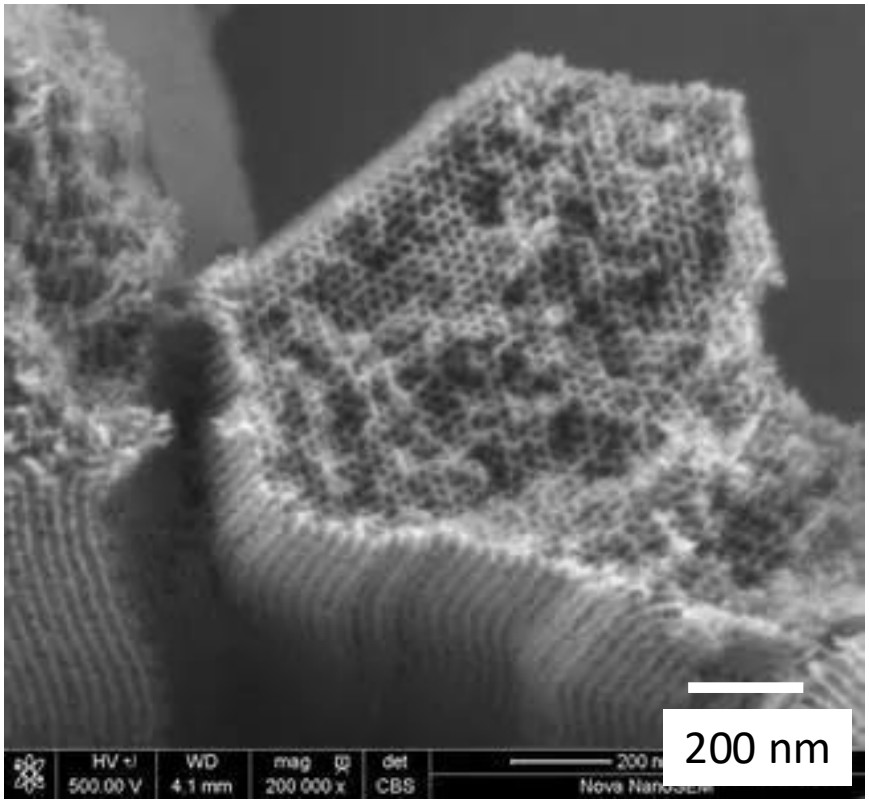
- Active Catalyst only
- Support + “Active” Catalyst



Catalysts Supports



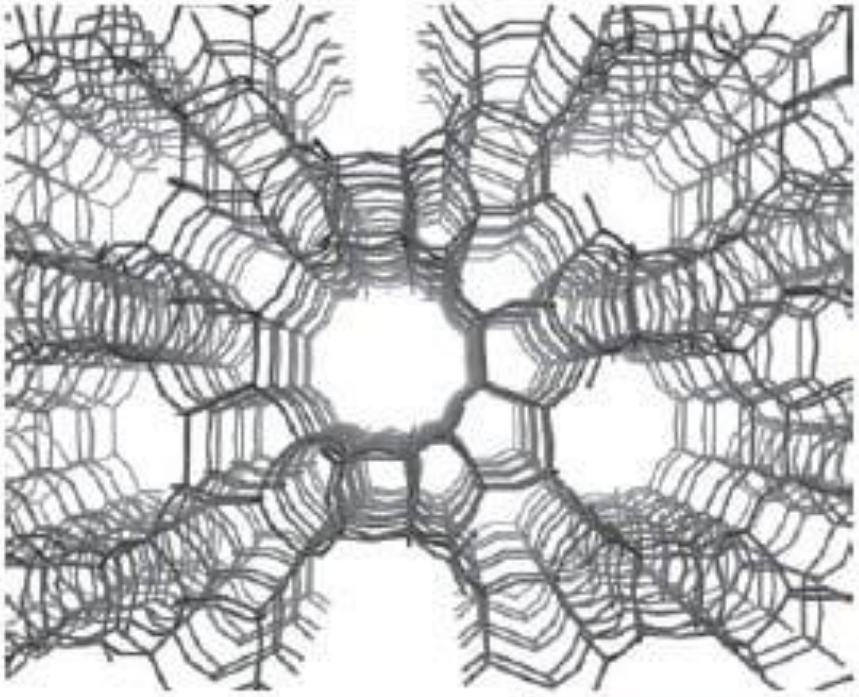
Zeolites

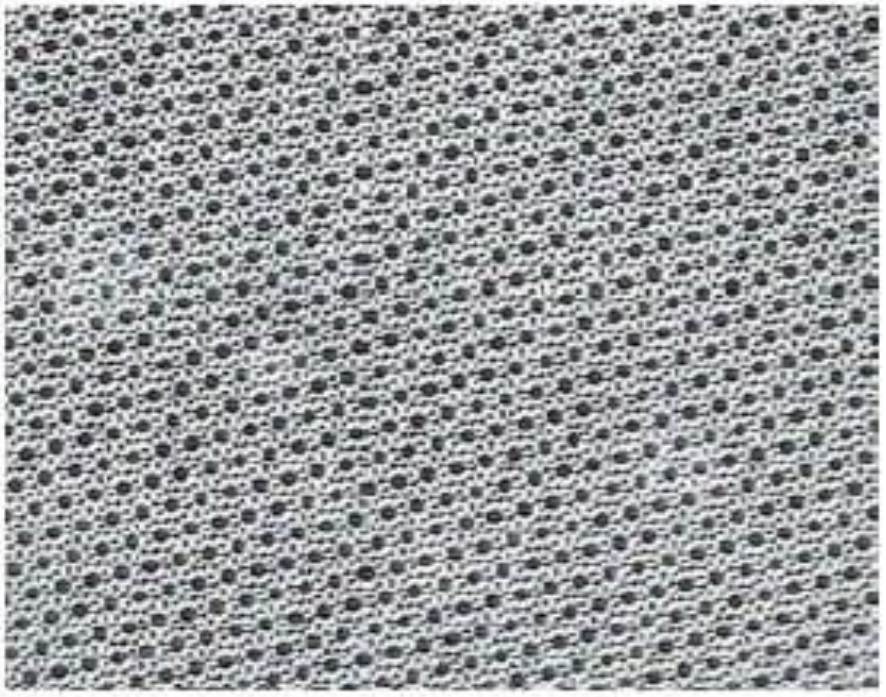


SBA-15

https://www.clariant.com/en/Innovation/Innovation-Spotlight-Videos/Zeolite-Catalyst-Technology

Zeolites





Examples of Catalytic Reactions

Catalytic process	Gas/vapor composition	Catalysts
Auto emissions control	N_2, O_2, HCs, CO, NO	Pt-Rh/Al_2O_3
Ammonia synthesis and regeneration	H_2, N_2	Fe/K/Al_2O_3
	Traces O_2, H_2O	
Catalytic cracking	HCs, H_2, H_2O	LaY-zeolite
CO oxidation, gas turbine exhaust	N_2, O_2, 400 ppm CO, 100–400 ppm SO_2	Pt/Al_2O_3
Fischer–Tropsch	CO, H_2, H_2O, CO_2, HCs	Fe/K/Cu/SiO_2
Fischer–Tropsch	CO, H_2, H_2O, HCs	Co/SiO_2
Steam reforming and regeneration in H_2O	CH_4, H_2O, CO H_2, CO_2	Ni/Al_2O_3

Catalyst Deactivation (P707)

- When the kinetics are separable…

$$r_A'' = a(\text{catalyst history}) \times r_A'(\text{fresh catalyst})$$

…it is possible to study *catalyst decay* and reaction kinetics independently.

- When the kinetics are non-separable…

$$r_A'' = r_A'(\text{catalyst history},\ \text{reaction kinetics})$$

Catalyst Deactivation

- separable kinetics

$$r_A' = a(\text{catalyst history}) \times r_A'(\text{fresh catalyst})$$

$$a(t) = r_A'(t) / r_A'(t_0) \qquad -r_A' = a(t) k(T) \times f(C_i)$$

  - Commercial reactors maintain constant production rate by increasing T (reaction rate constant increases), as catalyst decays (catalyst activity **a** decreases).

- experimental analysis of the decay rate is as:

$$r_d = \frac{-da}{dt} = p(a) \times k_d(T) \times h(C_i)$$

Catalyst Deactivation: Thermal Degradation

Page 711

- Sintering (aging)
  - Activity loss by loss of active surface caused by prolonged exposure to elevated gas-phase reaction temperatures.
  - Mechanistically…
    - Crystal agglomeration/growth, reducing internal surface area accompanied by narrowing/blocking of pore cross section.
    - Loss of support area due to pore collapse of active phase
    - Change in surface structure through recrystallization or other modes of defect elimination (active site loss).
  - Typically a 2nd order process;

$$r_d = -\frac{da}{dt} = k_d a^2$$

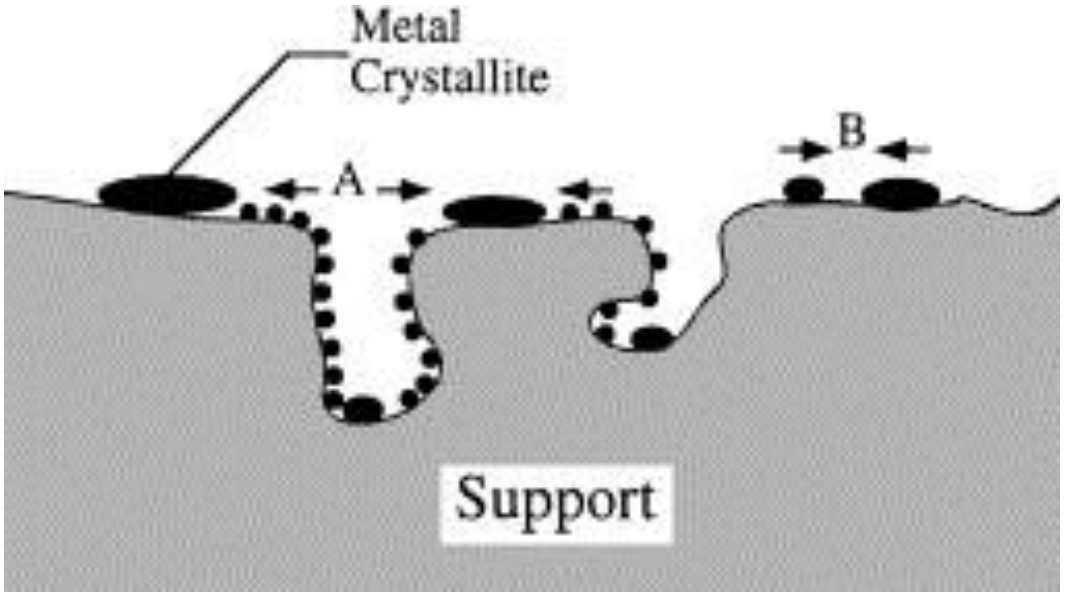
$$\int_1^a a^{-2} da = -k_d \int_0^t dt$$

$$a(t) = \frac{1}{1 + k_d t}$$

$$k_d = k_d(T_0)\exp\left[E/R\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$$

Thermal Degradation

Two conceptual models for crystallite growth due to sintering: (A) atomic migration or (B) crystallite migration.

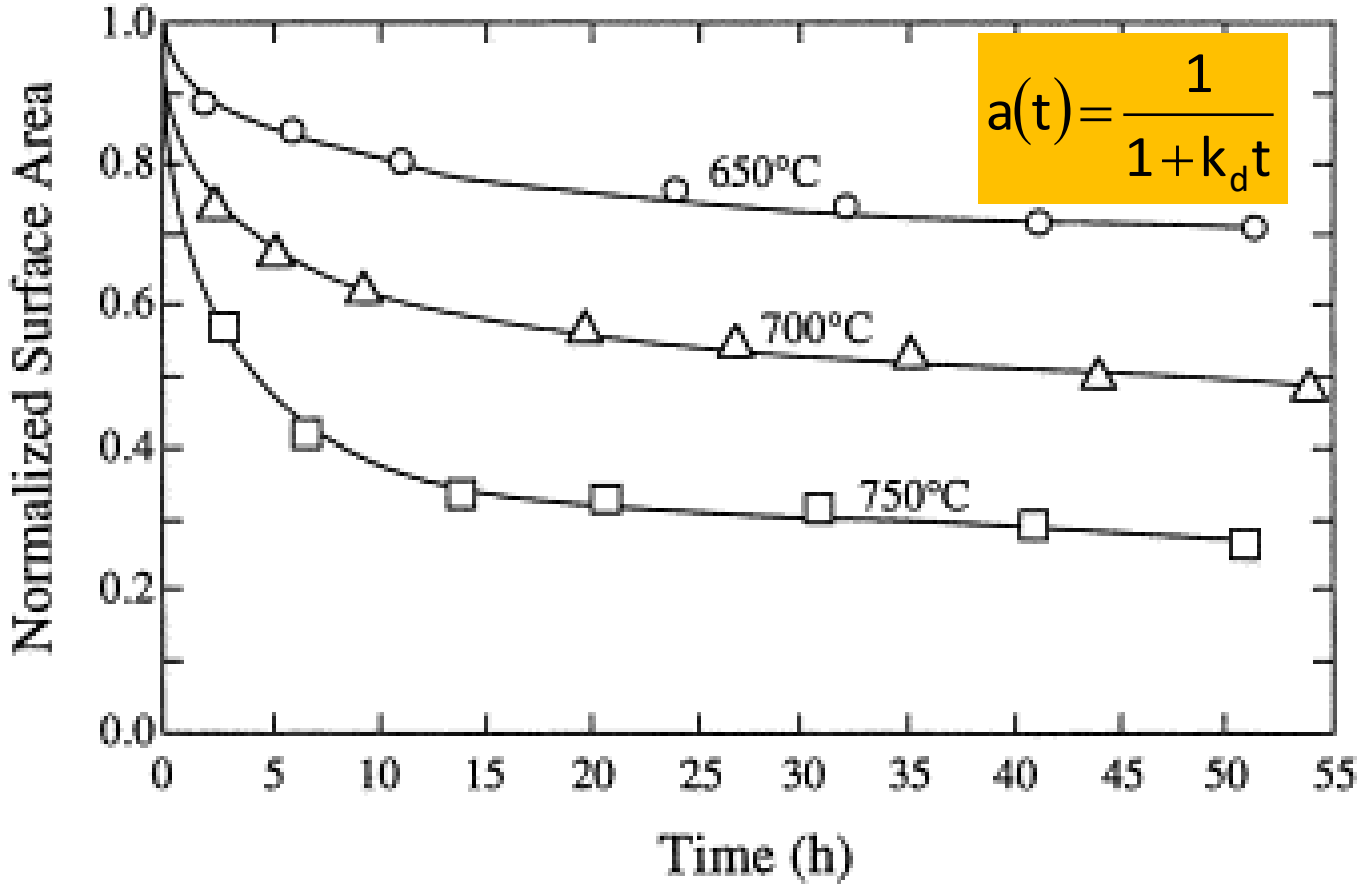


In general, sintering processes are kinetically slow (at moderate reaction temperatures) and irreversible or difficult to reverse. Thus, sintering is more easily prevented than cured.

Effects of important reaction and catalyst variables on sintering rates of supported metals

Variable	Effect
Temperature	Sintering rates are exponentially dependent on T; E_{act} varies from 30 to 150 kJ/mol; E_{act} decreases with increasing metal loading; it increases in the following order with atmosphere: NO, O_2, H_2, N_2
Atmosphere	Sintering rates are much higher for noble metals in O_2 than in H_2 and higher for noble and base metals in H_2 relative to N_2; sintering rate decreases for supported Pt in atmospheres in the following order: NO, O_2, H_2, N_2
Metal	Observed order of decreasing thermal stability in H_2 is Ru > Ir ≅ Rh > Pt; thermal stability in O_2 is a function of (1) volatility of metal oxide and (2) strength of metal oxide–support interaction
Support	Metal–support interactions are weak (bond strengths of 5–15 kJ/mol); with a few exceptions, thermal stability for a given metal decreases with support in the following order: Al_2O_3 > SiO_2 > carbon
Promoters	Some additives decrease atom mobility, e.g. C, O, CaO, BaO, CeO_2, GeO_2; others increase atom mobility, e.g. Pb, Bi, Cl, F, or S; oxides of Ba, Ca, or Sr are “trapping agents” that decrease sintering rate
Pore size	Sintering rates are lower for porous vs. non-porous supports; they decrease as crystallite diameters approach those of the pores

T effect on Catalyst Surface Area



Normalized nickel surface area (based on H_2 adsorption) vs. time data during sintering of 13.5% Ni/SiO_2 in H_2 at 650, 700 and 750°C

J. Catal., 81 (1983), p. 131

Catalyst Deactivation

- Poisoning
  - Mechanism involving the irreversible chemisorption of "poison" molecules to active sites.
  - Poisons may be reactants, products or impurities (sulfur is a common petrochemical processing catalyst poison).

A + S ↔ A•S

A•S ↔ B•S

B•S ↔ B + S

P + S → P•S

$$-r_A' = a(t) \times \left(\frac{kP_A}{1 + K_A P_A + K_B P_B} \right) = aP_A^n$$

$$r_d = -\frac{da}{dt} = k_d P_P^n a^q \qquad = k_d' a^q$$

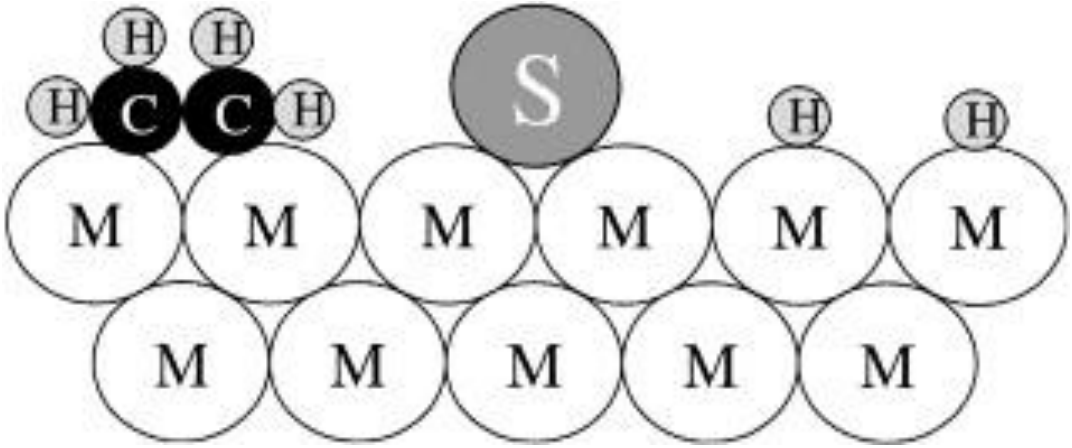
constant impurity partial P

upon integration

$$a = \left[1 - (1-q) k_d t\right]^{1/(1-q)}$$

Catalyst Deactivation: Poisoning

Conceptual model of poisoning by sulfur atoms of a metal surface during ethylene hydrogenation.



- Poisoning is the strong chemisorption of reactants, products or impurities on sites otherwise available for catalysis.
- Poisoning has operational meaning; whether a species acts as a poison depends upon its adsorption strength relative to the other species competing for catalytic sites.
- Oxygen can be a reactant in partial oxidation of ethylene to ethylene oxide on a silver catalyst and a poison in hydrogenation of ethylene on nickel.
- In addition to physically blocking of adsorption sites, adsorbed poisons may induce changes in the electronic or geometric structure of the surface.

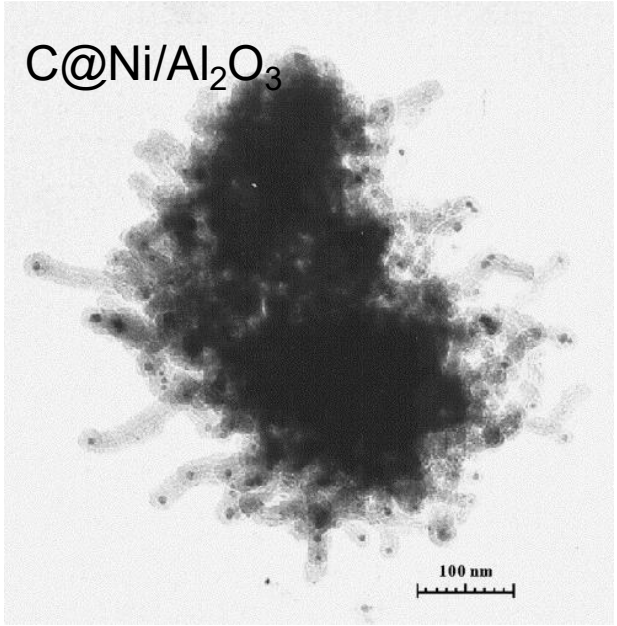
Poisons for selected catalysts in important representative reactions

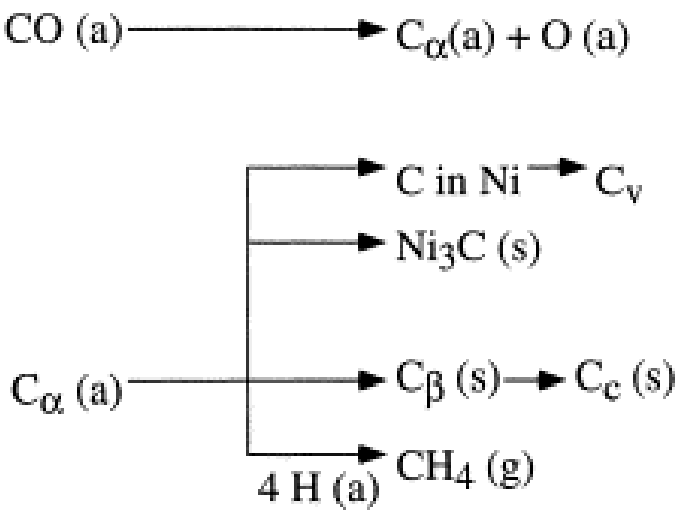
Catalyst	Reaction	Poisons
Silica-alumina, zeolites	Cracking	Organic bases, hydrocarbons heavy metals
Nickel, platinum, palladium	Hydrogenation dehydrogenation	Compounds of S, P, As, Zn, Hg, halides, Pb, NH_3, C_2H_2
Nickel	Steam reforming of methane, naphtha	H_2S, As
Iron, ruthenium	Ammonia synthesis	O_2, H_2O, CO, S, C_2H_2, H_2O
Cobalt, and iron	Fischer–Tropsch synthesis	H_2S, COS, As, NH_3, metal carbonyls
Noble metals on zeolites	Hydrocracking	NH_3, S, Se, Te, P
Silver	Ethylene oxidation to ethylene oxide	C_2H_2
Vanadium oxide	Oxidation selective catalytic reduction	As, Fe, K, Na from fly ash
Platinum, palladium	Oxidation of CO and hydrocarbons	Pb, P, Zn, SO_2, Fe
Cobalt and molybdenum sulfides	Hydrotreating of residue	Asphaltenes, N compounds, Ni, V

Common poisons classified according to chemical structure

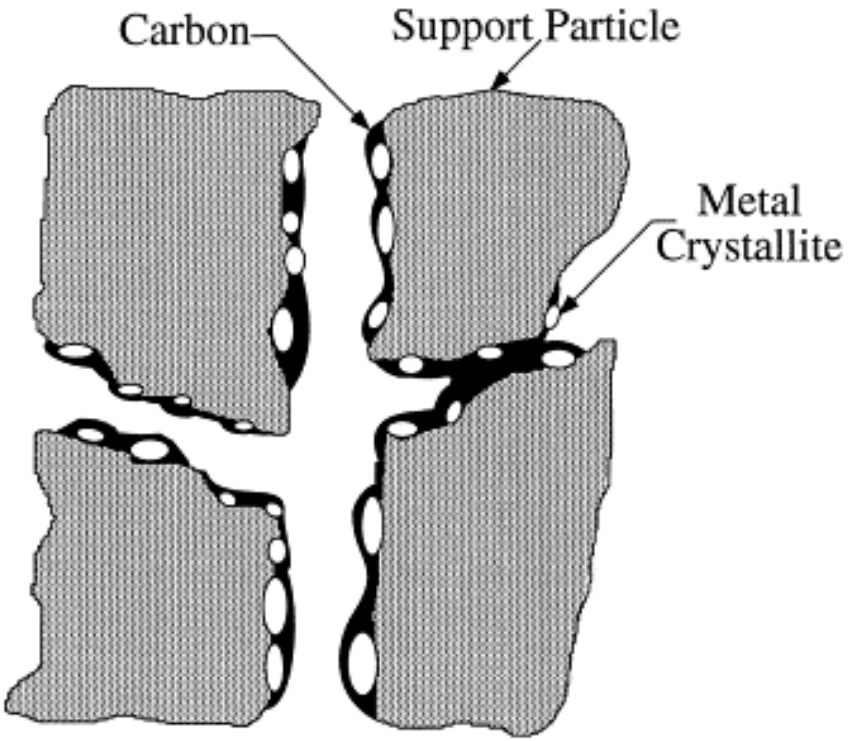
Chemical type	Examples	Type of interaction with metals
Groups VA and VIA	N, P, As, Sb, O, S, Se, Te	Through s- and p-orbitals; shielded structures are less toxic
Group VIIA	F, Cl, Br, I	Through s- and p-orbitals; formation of volatile halides
Toxic heavy metals and ions	As, Pb, Hg, Bi, Sn, Zn, Cd, Cu, Fe	Occupy d-orbitals; may form alloys
Molecules which adsorb with multiple bonds	CO, NO, HCN, benzene, acetylene, other unsaturated hydrocarbons	Chemisorption through multiple bonds and back bonding

Catalyst Deactivation: Fouling/coking





Fouling/Coking



Conceptual model of fouling, crystallite encapsulation and pore plugging of a supported metal catalyst due to carbon deposition.

Catalyst Deactivation: Fouling/coking

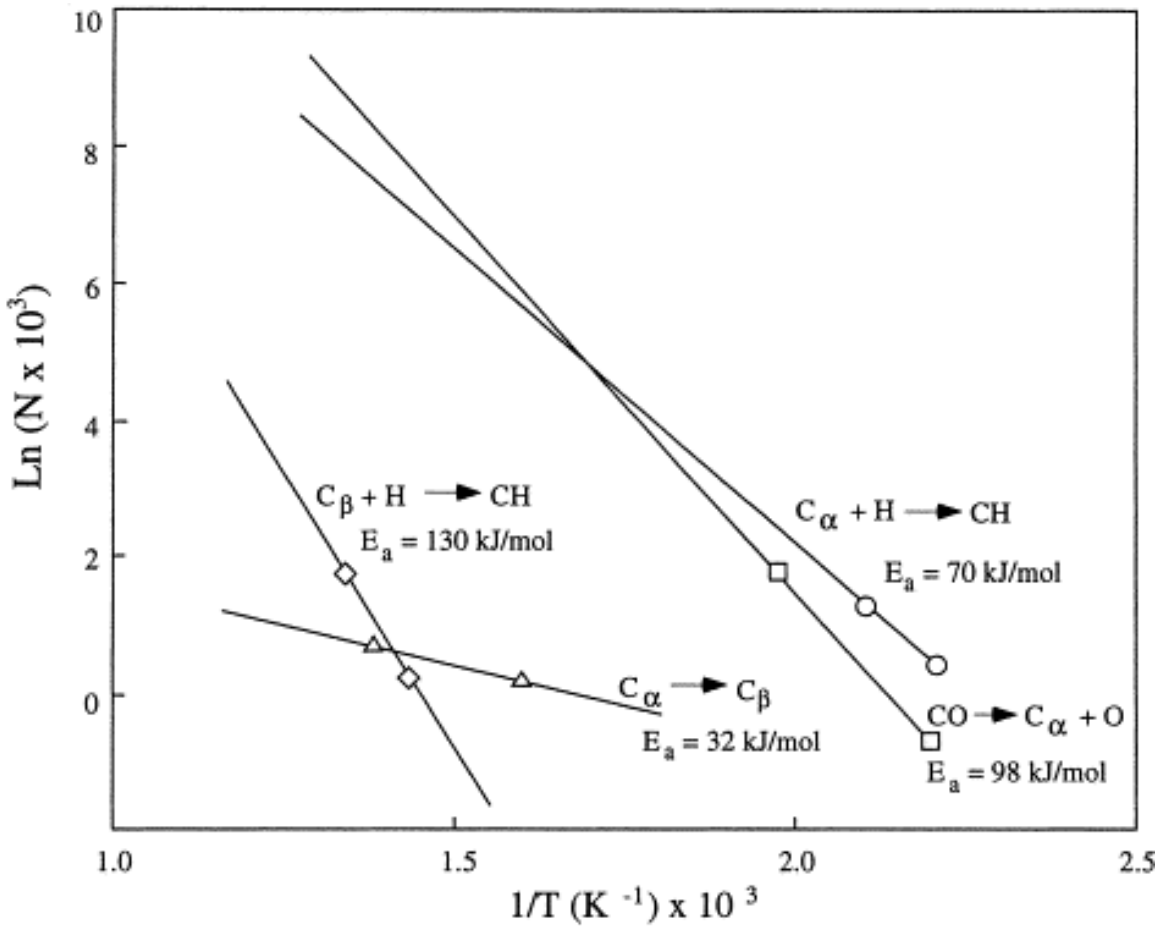
- Fouling/Coking
  - Deposition of carbonaceous material on catalyst surface
  - Catalyst activity level is a function of the amount of carbon deposited on the catalyst surface (C_c): $C_c = At^n$

    where *A* and *n* are fouling parameters dependent on the type of gas being processed.
  - Activity is expressed as $f(C_c)$ by one of the following:

$$a(t) = \frac{1}{1 + C_c^p} = \frac{1}{1 + A^p t^{np}} \qquad a = e^{-\alpha_1 C_c} \qquad a = \frac{1}{1 + \alpha_2 C_c}$$

$$a(t) = \frac{1}{1 + k_d t}$$

Rate of C_x Formation



Rates of formation (ln scale) and hydrogenation of C_α and C_β *vs.* reciprocal temperature

Catal. Rev.-Sci. Eng., 24 (1982), p. 67

Coking

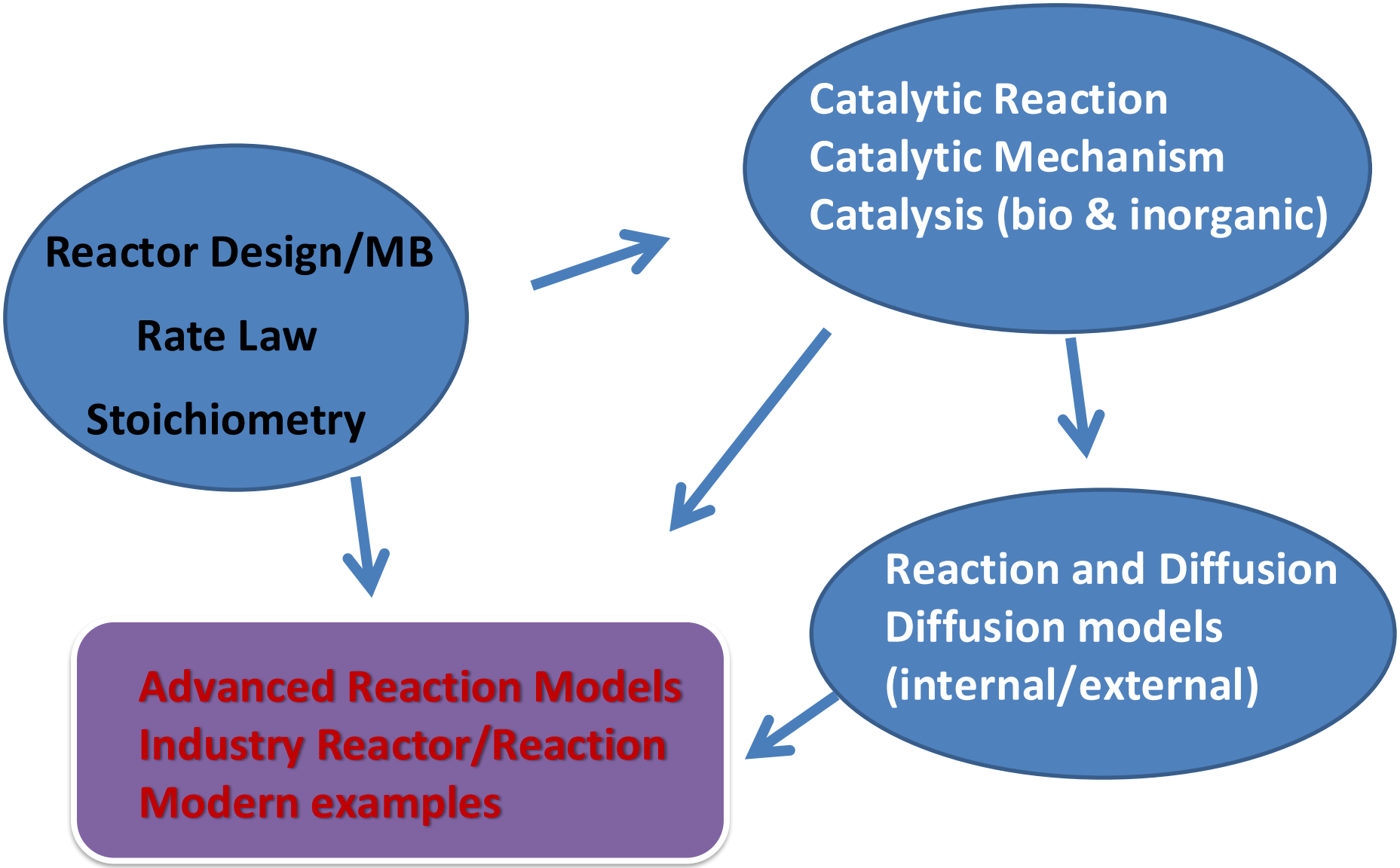
- Coking is not understood mechanistically.
- Coking essentially has two effects on the catalyst
  - Site coverage (coke covers the active catalytic site)
  - Pore blockage (coke creates diffusional limitations by changing pore size distribution)
    - The growth of coke in a pore network leads to a probabilistic formulation of the problem, by which the probability that a site is accessible (P) is related to a probability density q(D), which is expressed in terms of the pore network structure

Anything good? Insights?

Examples of reactions of gases/vapors with catalytic solids to produce inactive phases

Catalytic process	Gas/vapor composition	Catalytic solid	Deactivating chemical reaction
Auto emissions control	N_2, O_2, HCs, CO, NO	Pt-Rh/Al_2O_3	$2Rh_2O_3 + \gamma\text{-}Al_2O_3 \rightarrow RhAl_2O_4 + 0.5O_2$
Ammonia synthesis and regeneration	H_2, N_2	Fe/K/Al_2O_3	Fe → FeO at >50 ppm O_2
	Traces O_2, H_2O		Fe → FeO at $H_2O/H_2 > 0.16$
Catalytic cracking	HCs, H_2, H_2O	LaY-zeolite	H_2O induced Al migration from zeolite framework causing zeolite destruction
CO oxidation, gas turbine exhaust	N_2, O_2, 400 ppm CO, 100–400 ppm SO_2	Pt/Al_2O_3	$2SO_3 + \gamma\text{-}Al_2O_3 \rightarrow Al_2(SO_4)_3$ which blocks catalyst pores
Fischer–Tropsch	CO, H_2, H_2O, CO_2, HCs	Fe/K/Cu/SiO_2	$Fe_5C_2 \rightarrow Fe_3O_4$ due to oxidation at high X_{CO} by products H_2O, CO_2
Fischer–Tropsch	CO, H_2, H_2O, HCs	Co/SiO_2	$Co + SiO_2 \rightarrow CoO{\cdot}SiO_2$ and collapse of SiO_2 by product H_2O
Steam reforming and regeneration in H_2O	CH_4, H_2O, CO H_2, CO_2	Ni/Al_2O_3	$Ni + Al_2O_3 \rightarrow Ni_2Al_2O_4$

Our Scope:


Reactor Design/MB
Rate Law
Stoichiometry
Catalytic Reaction
Catalytic Mechanism
Catalysis (bio & inorganic)
Reaction and Diffusion
Diffusion models
(internal/external)
Advanced Reaction Models
Industry Reactor/Reaction
Modern examples